

Co-working Space Desk & Room Booking System

Overview

Build a system where members book desks/meeting rooms for time slots, and admins manage inventory, approve/reject bookings, and prevent double-booking conflicts.

User Roles

- **Visitor:** Browse available spaces (desks/rooms), view details, view real-time availability calendar.
- **Member:** Register/login, book a desk/room for a date+time slot, view/cancel own bookings.
- **Admin:** Manage spaces (CRUD), view all bookings, approve/reject booking requests, block out maintenance slots.

Features

Visitor

- List all spaces (desks, meeting rooms) with capacity, type, amenities.
- Search by space name or type.
- Filter by space type, capacity, date availability.

- View space details with availability calendar for a given date.
- Pagination on space list.

Member

- Register/Login (JWT).
- Book a space for a specific date + start/end time slot.
- **Cannot** book a slot that overlaps an existing approved/pending booking for that space (concurrency-safe - two members hitting "book" at the same instant for the same slot must not both succeed).
- View own bookings with status (pending/approved/rejected/cancelled).
- Cancel own pending or approved booking (future bookings only).

Admin

- Add/Edit/Delete spaces.
- Block out maintenance windows (space unavailable for a date/time range).
- View all bookings, filterable by status/date/space.
- Approve or reject pending bookings.
- Auto-reject any other pending booking that overlaps once one is approved.

Technical Requirements

Backend

- Node.js + Express.
- JWT auth with role-based authorization (member/admin). Include token expiry + refresh token flow (not just a long-lived access token).
- REST API for CRUD + booking workflow.

- **Concurrency handling on booking creation** - use DB-level constraint, transaction, or locking so overlapping bookings can't both be created under simultaneous requests.
- Centralized error-handling middleware, consistent JSON error shape.
- Input validation on all write endpoints (reject malformed/overlapping/past-dated bookings).
- Rate limiting on login/register endpoints.
- Indexes on fields used for search/filter/date-range queries.

Frontend

- React + Axios.
- Public space listing: search, filters, pagination.
- Availability calendar/view for a selected space (any charting/calendar lib or custom).
- Member dashboard: bookings list + cancel action.
- Admin dashboard: space CRUD, booking approval queue.
- Responsive UI.

Database

- MongoDB or PostgreSQL - either is fine.

Bonus (Optional - do not let this eat core time)

- Docker + docker-compose (`docker-compose up` brings up backend + frontend + DB, no manual setup) - plus point for fast reviewer onboarding.
 - Email/notification stub on booking status change.
 - Deployed link (frontend + backend).
 - Swagger/Postman collection.
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Deliverables

- GitHub repo with README.
- `.env.example` file.
- `docker-compose.yml` for one-command local run (optional, plus point).
- Postman/Swagger API docs (optional)